

Bayesian data fusion developments for causal inference

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SUMMARY

Combining randomised controlled trials with observational data for causal inference has attracted substantial attention among researchers over the past two decades. While randomised trials remain the gold standard in evidence-based medicine, their modest sample sizes limit efficiency, whereas naive pooling with observational data introduces confounding bias. A foundational Bayesian framework for data fusion is the power prior method of Ibrahim & Chen (2000). However, standard formulations face practical challenges in the specification of the power hyperparameter and the lack of explicit mechanisms to measure data commensurability. This paper reviews some recent Bayesian methodologies which address these limitations, ranging from power prior extensions and robust meta-analytic predictive priors to power likelihood approaches.

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Some key words: Bayesian statistics; Causal inference; Data fusion; Clinical Trial; Power Prior; Meta-analytic-predictive prior; Power likelihood.

1. INTRODUCTION

Evaluating interventions in the biomedical sciences mostly requires addressing causal rather than associational questions, seeking to determine how outcomes change under specific treatments. In this context, causal inference provides the formal mathematical framework to distinguish cause-and-effect relationships from mere statistical associations (Rubin, 1974; Pearl, 2009).

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Experimental data, such as randomised controlled trials (RCTs), serve as the gold standard for answering causal questions by eliminating confounding through design. However, RCTs are frequently constrained by small sample sizes, high operational costs, lengthy timelines, and restrictive eligibility criteria. Conversely, observational real-world data (RWD) provide large cohorts, though they are vulnerable to unmeasured confounding and selection bias.

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Combining both sources via *data fusion* (Bareinboim & Pearl, 2016) has thus emerged as an attractive strategy that balances the internal validity of RCTs with the statistical power and generalisability of RWD. Although these methodologies generalise beyond clinical trials, RCT augmentation provides the primary motivating framework. In this setting, the Bayesian paradigm offers a natural foundation to introduce external evidence into prior distributions.

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A core precursor to this literature is the class of *power prior distributions* of Ibrahim & Chen (2000). Although originally proposed for general regression settings, it aligns naturally with the goal of augmenting RCTs with external RWD, serving as the baseline mathematical engine for several subsequent methodologies. While Ibrahim & Chen (2000) used *historical* data to refer to information arising from previous similar studies, we use *historical*, *external*, and RWD interchangeably as auxiliary information for the target study.

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Following the notation of the power prior approach, given $a_0 \in [0, 1]$, the power prior distribution for a parameter θ is defined as

$$\pi(\theta \mid D_0, a_0) \propto L(\theta \mid D_0)^{a_0} \pi_0(\theta \mid c_0), \quad (1)$$

where $D_0 = (n_0, y_0, X_0)$ denotes historical data, comprising sample size n_0 , response vector $y_0 \in \mathbb{R}^{n_0}$, and covariate matrix $X_0 \in \mathbb{R}^{n_0 \times p}$. The function $L(\theta \mid D_0)$ denotes the historical likelihood, and $\pi_0(\theta \mid c_0)$ is the initial prior, usually non-informative, indexed by a specified hyperparameter vector c_0 . When current experimental data $D_e = (n_e, y_e, X_e)$ with likelihood $L(\theta \mid D_e)$ become available, updating (1) via Bayes' theorem yields the posterior distribution

$$\pi(\theta \mid D_e, D_0, a_0) \propto L(\theta \mid D_e) L(\theta \mid D_0)^{a_0} \pi_0(\theta \mid c_0). \quad (2)$$

In this formulation, a_0 acts as a precision parameter that weights the influence of historical data on the current study. Tuning $a_0 \in (0, 1)$ disperses prior concentration around historical point estimates, providing robustness against prior–data conflict. Setting $a_0 = 1$ recovers the standard Bayesian update; when $a_0 = 0$, the external evidence is excluded, yielding $\pi(\theta \mid D_0, a_0 = 0) = \pi_0(\theta \mid c_0)$. Posterior computation is generally straightforward using standard Markov chain Monte Carlo algorithms.

In clinical trials, borrowing external real-world information via power priors narrows posterior credible intervals while improving statistical power. Furthermore, constructing priors directly from D_0 facilitates semi-automatic prior elicitation, making the approach advantageous for model and variable selection.

Despite these benefits, the classical implementation presents two notable challenges. First, when a_0 is treated as known, its elicitation relies heavily on subjective judgment, which can introduce arbitrary bias. While Ibrahim & Chen (2000) also considered a hierarchical specification treating a_0 as random, their original joint power prior formulation for (θ, a_0) violates the likelihood principle. Second, the baseline formulation lacks an internal mechanism to evaluate the empirical compatibility—or *commensurability*—between D_0 and D_e .

This paper reviews the methodological advancements overcoming these limitations. Section 2 summarises power parameter normalisation and its extensions (Duan et al., 2006; Neuenschwander et al., 2009). Section 3 explores meta-analytic-predictive priors (Schmidli et al., 2014) and two-source hierarchical shrinkage estimation (Röver & Friede, 2020). Section 4 examines power likelihood methods combined with the frugal parameterisation (Evans & Didelez, 2024; Lin et al., 2026).

2. EXTENSIONS OF THE POWER PRIOR METHOD

2.1. The modified power prior

For ease of exposition, consider first the one-historical-study setting (D_0) before generalising to multiple cohorts in Section 2.3. When the discounting parameter a_0 is treated as random, uncertainty is modelled by specifying an independent hyperprior $\pi(a_0)$ supported on $[0, 1]$, typically a Beta(α, β) or uniform distribution. Originally, Ibrahim & Chen (2000) formulated the joint power prior for (θ, a_0) as

$$\pi(\theta, a_0 \mid D_0) \propto L(\theta \mid D_0)^{a_0} \pi_0(\theta \mid c_0) \pi(a_0). \quad (3)$$

However, this specification exhibits some theoretical and practical drawbacks. First, it violates the likelihood principle: scaling $L(\theta \mid D_0)$ by an arbitrary positive constant alters the posterior distribution of (θ, a_0) . As shown by Duan et al. (2006) and Neuenschwander et al. (2009), the joint distribution in (3) decomposes into $\pi(\theta, a_0 \mid D_0) = \pi(\theta \mid D_0, a_0) \pi(a_0)$, and so requires retaining the

normalising constant in (1) depending on a_0 , $C(a_0) = [\int_{\Theta} L(\theta | D_0)^{a_0} \pi_0(\theta | c_0) d\theta]^{-1}$. Omitting $C(a_0)$ leads to severe artefacts: standard marginalisation $\pi(\theta | D_0) = \int \pi(\theta | D_0, a_0) \pi(a_0) da_0$ fails, and the posterior marginal distribution $\pi(a_0 | D_e, D_0)$ tends to collapse towards zero, under-borrowing historical information even when the historical and current trial data are fully identical.

To restore standard probability calculus, Duan et al. (2006) and Neuenschwander et al. (2009) proposed the *modified power prior*:

$$\pi(\theta, a_0 | D_0) = C(a_0) L(\theta | D_0)^{a_0} \pi_0(\theta | c_0) \pi(a_0), \quad (4)$$

defined on the subset $A = \{a_0 \in [0, 1] : 0 < \int_{\Theta} L(\theta | D_0)^{a_0} \pi_0(\theta | c_0) d\theta < \infty\}$.

Under an improper initial prior $\pi_0(\theta)$, the domain A explicitly constrains a_0 to guarantee proper power priors (e.g., $a_0 > 1/n_0$ under the standard reference prior for Gaussian models; Duan et al., 2006).

2.2. Properties and posterior inference

Upon observing current trial data D_e , the resulting joint posterior becomes

$$\pi(\theta, a_0 | D_e, D_0) \propto L(\theta | D_e) L(\theta | D_0)^{a_0} \pi_0(\theta | c_0) C(a_0) \pi(a_0) I(a_0 \in A), \quad (5)$$

yielding the marginal posterior distributions for the discounting parameter and the target parameter:

$$\pi(a_0 | D_e, D_0) \propto \pi(a_0) C(a_0) \int_{\Theta} L(\theta | D_e) L(\theta | D_0)^{a_0} \pi_0(\theta | c_0) d\theta, \quad (6)$$

$$\pi(\theta | D_e, D_0) \propto L(\theta | D_e) \pi_0(\theta | c_0) \int_A C(a_0) L(\theta | D_0)^{a_0} \pi(a_0) da_0. \quad (7)$$

Under (4), the power parameter adapts to empirical compatibility (Duan et al., 2006). In conjugate settings, when D_0 and D_e are concordant, the posterior mode of a_0 approaches 1 as sample size grows, maximising precision; conversely, severe conflict forces a_0 towards 0 (Duan et al., 2006; Neuenschwander et al., 2009). Nevertheless, estimating a_0 remains challenging in practice. As Neuenschwander et al. (2009) noted, a single historical study alongside the current trial provides very limited information on between-trial heterogeneity, leaving the posterior of a_0 sensitive to its prior unless sample sizes are large. This difficulty in learning a_0 from two sources motivates alternative hierarchical borrowing structures.

2.3. Extension to multiple external cohorts

When synthesising H historical studies, $D_0 = (D_{01}, \dots, D_{0H})$, the formulation generalises by assigning study-specific borrowing weights $a_0 = (a_{01}, \dots, a_{0H})^\top \in [0, 1]^H$ (Duan et al., 2006):

$$\pi(\theta, a_0 | D_0) = \frac{\prod_{h=1}^H L(\theta | D_{0h})^{a_{0h}} \pi_0(\theta | c_0)}{\int_{\Theta} \prod_{h=1}^H L(\theta | D_{0h})^{a_{0h}} \pi_0(\theta | c_0) d\theta} \prod_{h=1}^H \pi(a_{0h}), \quad (8)$$

where each $a_{0h} \sim \text{Beta}(\alpha_h, \beta_h)$ independently evaluates the commensurability of source h .

While $C(a_0)$ is analytically tractable in conjugate models, non-conjugate and general regression settings require numerical integration or Markov chain Monte Carlo algorithms (Neuenschwander et al., 2009). In the multi-cohort setting ($H \geq 2$), posterior inference and the theoretical properties established for the single-source case carry over directly, with each component a_{0h} adapting dynamically to the empirical compatibility between historical study h and the current trial.

3. HIERARCHICAL AND META-ANALYTIC SHRINKAGE PRIORS

3.1. *Meta-analytic-predictive (MAP) priors*

An alternative paradigm for resolving multi-source heterogeneity and parameterising commensurability is the robust meta-analytic-predictive (MAP) prior framework (Schmidli et al., 2014). Rather than relying on fractional likelihood discounting, this approach constructs a hierarchical predictive prior for the trial control parameter, which is approximated by a mixture of conjugate densities. This selective borrowing targets the control response parameter alone, enabling researchers to downsize the concurrent control arm without altering the experimental group.

Consider H historical control cohorts with summary outcomes $Y_h \sim F(\psi_h; n_h)$ for $h = 1, \dots, H$, alongside a target trial with control outcome $Y_* \sim F(\psi_*; n_*)$. Schmidli et al. (2014) formulate the method for $F(\cdot)$ within the regular one-parameter exponential family, where sample sizes n_h and totals Y_h serve as sufficient statistics. Although originally framed for aggregate summary data, this hierarchical framework extends naturally to individual patient-level data via meta-regression to adjust for baseline covariate imbalances.

On a canonical link scale $\theta = g(\psi)$ (e.g., the logit link for binomial proportions or identity link for Gaussian means), similarity across historical and new trials is formalised through parameter exchangeability:

$$\theta_1, \dots, \theta_H, \theta_* \mid \mu, \tau \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \tau^2), \quad \pi(\mu, \tau) = \pi(\mu)\pi(\tau), \quad (9)$$

where conditionally on (μ, τ) the parameters are mutually independent, μ denotes the population mean, and τ quantifies between-trial heterogeneity. While a diffuse prior suffices for μ , the heterogeneity parameter τ requires regularising hyperpriors—typically half- t , half-normal, or half-Cauchy distributions calibrated to penalise implausibly large variance components (Gelman, 2006; Polson & Scott, 2012). Small values of τ reflect cross-trial concordance and facilitate borrowing, whereas larger values widen the predictive prior and attenuate historical influence. Crucially, joint estimation across all sources is mathematically equivalent to deriving the MAP prior $\pi_{\text{MAP}}(\psi_*) = \pi(\psi_* \mid Y_1, \dots, Y_H)$ solely from historical data and subsequently updating it with new trial data Y_* via Bayes’ theorem (Schmidli et al., 2014).

Because the predictive density $\pi(\psi_* \mid Y_1, \dots, Y_H)$ lacks a closed analytical form, Schmidli et al. (2014) approximate the empirical Markov chain Monte Carlo predictive distribution via a K -component mixture of conjugate densities:

$$\hat{\pi}_{\text{MAP}}(\psi_*) = \sum_{k=1}^K w_k \pi_c(\psi_* \mid \xi_k), \quad \sum_{k=1}^K w_k = 1, \quad w_k > 0, \quad (10)$$

where ξ_k denotes the vector of component-specific hyperparameters (e.g., $\xi_k = (a_k, b_k)^\top$ for Beta(a_k, b_k) distributions in Bernoulli models). As established by Dalal & Hall (1983) and Diaconis & Ylvisaker (1985), finite conjugate mixtures can approximate arbitrary prior densities. The weights $\{w_k\}_{k=1}^K$ and hyperparameters $\{\xi_k\}_{k=1}^K$ are estimated by minimising the Kullback–Leibler divergence from the exact predictive density $\pi_{\text{MAP}}(\psi_*)$ to $\hat{\pi}_{\text{MAP}}(\psi_*)$, which is numerically equivalent to maximum likelihood estimation over the posterior predictive draws using the expectation–maximisation algorithm. Typically, setting $K \in \{2, 3\}$ yields an accurate fit while preserving analytical tractability. While the overarching predictive concept extends to non-exponential families or multi-parameter models, restricting the prospective sampling model to the one-parameter exponential family yields closed-form marginal likelihoods and posterior updates, bypassing numerical integration during trial analysis.

3.2. Robust MAP priors and posterior updating

Standard conjugate priors enforce a linear compromise between prior and sample means irrespective of data discordance. To protect against violations of the exchangeability assumption (9), Schmidli et al. (2014) construct the *robust MAP prior* by augmenting the informative mixture with a proper, weakly informative conjugate component $\pi_V(\psi_*)$ weighted by $w_R \in [0, 1]$: 155

$$\hat{\pi}_{HR}(\psi_*) = (1 - w_R) \sum_{k=1}^K w_k \pi_c(\psi_* | \xi_k) + w_R \pi_V(\psi_*). \quad (11)$$

The weight w_R reflects the prior probability of non-exchangeability between historical control data and the current trial, commonly fixed at $w_R \in \{0.10, 0.20\}$ in adherence to Cromwell's rule (Schmidli et al., 2014). Upon observing new control data Y_* , the posterior retains the conjugate mixture form: 160

$$\hat{\pi}_{HR}(\psi_* | Y_*) = (1 - \tilde{w}_R) \sum_{k=1}^K \tilde{w}_k \pi_c(\psi_* | \xi_k, Y_*) + \tilde{w}_R \pi_V(\psi_* | Y_*). \quad (12)$$

Integrating each prior component against the sampling density $f(Y_* | \psi_*)$ across the parameter space Ψ yields the marginal predictive likelihoods:

$$m_k = \int_{\Psi} f(Y_* | \psi_*) \pi_c(\psi_* | \xi_k) d\psi_*, \quad m_V = \int_{\Psi} f(Y_* | \psi_*) \pi_V(\psi_*) d\psi_*. \quad (13)$$

Normalising these predictive factors determines the updated posterior weight of the non-informative component $\tilde{w}_R$ and the sub-weights $\tilde{w}_k$ within the informative mixture: 165

$$\tilde{w}_R = \frac{w_R m_V}{w_R m_V + (1 - w_R) \sum_{j=1}^K w_j m_j}, \quad \tilde{w}_k = \frac{w_k m_k}{\sum_{j=1}^K w_j m_j}, \quad (14)$$

such that the posterior weight assigned to the k th informative component in (12) is given by $(1 - \tilde{w}_R) \tilde{w}_k$.

The ratio m_k/m_V acts as a Bayes factor comparing the k th informative component to the vague component: under prior–data concordance, the factors m_k dominate to facilitate information borrowing, whereas severe conflict ($m_k \ll m_V$) drives $\tilde{w}_R \rightarrow 1$, adaptively discounting historical evidence to guard against estimation bias. The prior effective sample size can be evaluated via the log-prior curvature at the mode following Morita et al. (2008), providing an operational baseline for prospective sample size reduction in trial design. 170

It is worth noting that the MAP formulation requires multiple historical studies ($H \geq 3$) to reliably estimate the between-study heterogeneity parameter τ . Indeed, when borrowing from a single external dataset ($H = 1$), τ^2 cannot be reliably learned from data alone, leaving posterior inferences sensitive to the chosen hyperprior. This limitation motivates alternative shrinkage models tailored specifically to two-source data fusion, as developed by Röver & Friede (2020). 175

3.3. Reference-based shrinkage for pairwise data fusion

When synthesising evidence from only two data sources ($H = 1$), such as augmenting a small RCT with a larger observational registry, classical frequentist random-effects estimators often fail because the between-study heterogeneity τ is frequently estimated as zero, underestimating variance uncertainty (Röver & Friede, 2020). To address this limitation, Röver & Friede (2020) proposed a Bayesian shrinkage estimation framework within the normal–normal hierarchical model (NNHM) that enables dynamic information borrowing while propagating heterogeneity uncertainty. 180

Let $\hat{\theta}_e \sim \mathcal{N}(\theta_e, \sigma_e^2)$ denote the primary trial (experimental) estimate and $\hat{\theta}_o \sim \mathcal{N}(\theta_o, \sigma_o^2)$ the observational summary, where θ_e and θ_o represent contrast-level treatment parameters (e.g., log-odds or log-hazard ratios) with known standard errors σ_e and σ_o . When combining experimental and observational evidence, assuming symmetric exchangeability around an artificial pooled mean parameter is conceptually difficult to justify. Instead, Röver & Friede (2020) formulate an asymmetric *reference model* that establishes the randomised controlled trial directly as the target parameter ($\theta_e = \mu$), with observational discrepancy captured via a zero-mean offset:

$$\theta_e = \mu, \quad \theta_o \mid \mu, \kappa \sim \mathcal{N}(\mu, \kappa^2), \quad (15)$$

where $\pi(\mu) \propto 1$ and $\kappa = \sqrt{2}\tau$ denotes the discrepancy standard deviation re-scaled relative to the standard NNHM heterogeneity parameter $\tau \sim \text{HN}(s)$. In this pairwise setting, the formulation yields posterior shrinkage estimates for θ_e identical to the symmetric model. Conditional on τ , the posterior expectation for the target parameter takes the precision-weighted form

$$E\{\theta_e \mid \hat{\theta}_e, \hat{\theta}_o, \tau\} = \frac{\sigma_e^{-2}\hat{\theta}_e + (\sigma_o^2 + 2\tau^2)^{-1}\hat{\theta}_o}{\sigma_e^{-2} + (\sigma_o^2 + 2\tau^2)^{-1}}. \quad (16)$$

Marginal inference proceeds by numerically integrating over the posterior density $\pi(\tau \mid \hat{\theta}_e, \hat{\theta}_o)$, which penalises excessive heterogeneity.

As discrepancy increases ($\tau^2 \rightarrow \infty$), the observational borrowing weight vanishes, adaptively shrinking the posterior estimate back to the unaugmented trial evidence $\hat{\theta}_e$. This hierarchical variance discounting contrasts conceptually with the power prior approach, where historical evidence is discounted by raising its likelihood to an exponent $a_0 \in [0, 1]$. In fact, conditional on τ , the shrinkage estimator in (16) is algebraically equivalent to a conjugate power prior model with discounting parameter fixed at $a_0 = \sigma_o^2 / (\sigma_o^2 + 2\tau^2)$, establishing a direct analytical connection between variance inflation and likelihood exponentiation.

Simulation results in Röver & Friede (2020) demonstrate that this framework behaves robustly across plausible clinical scenarios. Despite the inherent limitation of synthesising only two evidence sources, it achieves substantial precision gains while providing a reliable alternative for pairwise data fusion. Moreover, the framework extends naturally to multi-study syntheses, multi-level hierarchical structures (such as combining distinct meta-analyses), and the inclusion of study-level meta-regression covariates.

However, some methodological caveats apply. First, the proposed method relies fundamentally on the asymptotic normality of the contrast estimates, which may falter under sparse data or rare-event regimes. Second, synthesising observational evidence assumes rigorous pre-adjustment for confounding, as the symmetric offset variance accounts for dispersion rather than systematic directional bias. Finally, the formulation operates strictly on contrast-level aggregate summary statistics rather than individual patient data.

4. POWER LIKELIHOOD APPROACHES

4.1. Joint power likelihoods and frugal parameterisation

To overcome the challenges of hyperparameter specification and reliance on aggregate summary data, Lin et al. (2026) introduced an adaptive data-fusion framework combining joint power likelihoods, frugal parameterisations (Evans & Didelez, 2024), and data-adaptive predictive density optimisation. The aim is to augment a randomised controlled trial with large-scale observational evidence at the individual participant level to improve the precision of heterogeneous treatment effect estimation while controlling confounding.

Rather than converting auxiliary observational data into a static prior distribution, the approach specifies a joint power likelihood over both domains. Let $X_e = \{X_i : i \in D_e\}$ and $X_o = \{X_i : i \in D_o\}$ denote the observed sample collections from the experimental trial D_e of sample size n_e and the observational cohort D_o of sample size n_o , respectively. Let $S_i \in \{e, o\}$ denote the domain indicator, where $S_i = e$ indicates enrolment in the experimental trial ($i \in D_e$) and $S_i = o$ denotes membership in the observational cohort ($i \in D_o$). For each unit i , $X_i = (Y_i, T_i, W_i)$, where $Y_i \in \mathbb{R}$ is the observed outcome, $T_i \in \{0, 1\}$ is the binary treatment indicator, and $W_i \in \mathbb{R}^d$ is the vector of pre-treatment baseline covariates. 230

Under unit-level independence with densities $f_s(X_s; \phi) = \prod_{i \in D_s} f_s(X_i; \phi)$ for source $S \in \{e, o\}$, the joint posterior for the parameter vector $\phi \in \mathbb{R}^P$ with discounting rate $\eta \in [0, 1]$ is: 235

$$\pi(\phi \mid X_e, X_o, \eta) \propto f_e(X_e; \phi) \{f_o(X_o; \phi)\}^\eta \pi(\phi), \quad (17)$$

where $\pi(\phi)$ denotes the baseline prior. The vector $\phi = (\theta, \gamma, \psi) \in \mathbb{R}^P$ ($p = p_\theta + p_\gamma + p_\psi$) partitions the parameter space into shared causal parameters $\theta \in \mathbb{R}^{p_\theta}$ governing the treatment effect contrast, trial-specific nuisance parameters $\gamma \in \mathbb{R}^{p_\gamma}$, and observational nuisance parameters $\psi \in \mathbb{R}^{p_\psi}$ capturing background confounding and selection mechanisms. The parameter $\eta \in [0, 1]$ acts as a continuous discounting rate, where $\eta = 0$ yields the trial-only posterior $\pi(\phi \mid X_e)$ and $\eta = 1$ enforces unpenalised pooling. 240

To structure f_e and f_o without imposing restrictive distributional constraints on non-causal components, Lin et al. (2026) adopt the frugal parameterisation of marginal structural models (Evans & Didelez, 2024). Partitioning baseline covariates $W \in \mathbb{R}^d$ into effect modifiers $C \in \mathbb{R}^{d_c}$ ($C \subseteq W$) and remaining covariates $Z \in \mathbb{R}^{d-d_c}$ ($Z = W \setminus C$), the joint density for source $s \in \{e, o\}$ factorises as 245

$$f_s(y, t, w) = f_s(t, z \mid c) f_s(c) f_{Y(t) \mid C}^*(y \mid t, c; \theta) c_s^*(y, t, z \mid c; \zeta_s), \quad (18)$$

where $f_{Y(t) \mid C}^*$ represents the causal marginal structural model parameterised by θ and isolates the conditional average treatment effect $\tau(c)$, while the baseline margin $f_s(t, z \mid c) f_s(c)$ and conditional copula c_s^* absorb domain-specific nuisance features with $\zeta_e = \gamma \in \mathbb{R}^{p_\gamma}$ for D_e and $\zeta_o = \psi \in \mathbb{R}^{p_\psi}$ for D_o . 250

Valid causal identification and borrowing across domains rest on four conditions. First, experimental randomisation ensures trial ignorability and positivity, namely $Y(t) \perp\!\!\!\perp T \mid W, S = e$ for $t \in \{0, 1\}$ and $\text{pr}(T = 1 \mid W, S = e) \in (0, 1)$. Second, trial representativeness guarantees $E\{Y(1) - Y(0) \mid C = c, S = e\} = \tau(c)$, allowing the analysis to target efficiency gains rather than generalisation reweighting. Third, observational contrast transportability asserts that $E\{Y(1) - Y(0) \mid C = c, S = o\} = \tau(c)$, demanding concordance solely on the conditional treatment scale $\tau(c)$ rather than marginal baseline equality, accommodating discrepancies in absolute outcome rates across cohorts. Finally, potential outcome specification demands that the counterfactual distribution $Y(t) \mid C$ is correctly modelled by $f_{Y(t) \mid C}^*(y \mid t, c; \theta)$; this restricts only the causal mechanism, leaving the observational nuisance distributions ψ unconstrained and protecting θ from observational model misspecification. 255
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4.2. Robust selection of the influence factor η

Following Lin et al. (2026), η is adaptively determined by evaluating out-of-sample predictive performance on the randomised trial data via the expected log predictive density (ELPD). Let $\mathcal{M}_\eta$ denote the power likelihood model indexed by η . The target utility is defined as the out-of-sample predictive density for new trial observations $X_e^* = (Y^*, T^*, W^*) \in \mathbb{R}^{d+2}$ sampled from the true 265

trial data-generating process g_e :

$$\text{ELPD}(\eta) = E_{X_e^* \sim g_e} \left\{ \log f_e(X_e^* | X_e, X_o, \eta) \right\}, \quad (19)$$

where $f_e(X_e^* | X_e, X_o, \eta) = \int_{\mathbb{R}^P} f_e(X_e^*; \phi) \pi(\phi | X_e, X_o, \eta) d\phi$ is the posterior predictive density under model $\mathcal{M}_\eta$. Because the true density g_e is unknown, Lin et al. (2026) propose to estimate (19) using K -fold cross-validation partitioned strictly over the trial sample D_e .

Let $\{D_e^{(k)}\}_{k=1}^K$ denote a partition of the trial indices into K folds, and write $D_e^{(-k)} = D_e \setminus D_e^{(k)}$ for the training folds. For each fold k , the model is fitted on the combined sample $\{D_e^{(-k)}, D_o\}$ with power η , and evaluated on the held-out trial fold $D_e^{(k)}$:

$$\widehat{\text{ELPD}}_{\text{cv}}(\eta) = \sum_{k=1}^K \sum_{i \in D_e^{(k)}} \log \left\{ \int_{\mathbb{R}^P} f_e(X_i; \phi) \pi(\phi | X_e^{(-k)}, X_o, \eta) d\phi \right\}. \quad (20)$$

The optimal influence factor is chosen over a discrete grid $\mathcal{H} \subset [0, 1]$:

$$\hat{\eta} = \arg \max_{\eta \in \mathcal{H}} \widehat{\text{ELPD}}_{\text{cv}}(\eta). \quad (21)$$

Evaluating the predictive criterion exclusively on the experimental data D_e preserves the internal validity of the randomised trial as the target benchmark. If the observational cohort introduces residual confounding or violates contrast transportability, the posterior predictive distribution under $\mathcal{M}_\eta$ diverges from the trial data-generating distribution on $D_e^{(k)}$, reducing $\widehat{\text{ELPD}}_{\text{cv}}(\eta)$ and shrinking $\hat{\eta} \rightarrow 0$. Conversely, when observational and trial causal mechanisms align, the variance reduction achieved increases the predictive density on held-out trial units, yielding $\hat{\eta} > 0$.

4.3. Asymptotic approximations, computation, and inference

By the Bernstein–von Mises theorem, as $n_e, n_o \rightarrow \infty$, the power likelihood posterior converges to a p -dimensional Gaussian distribution:

$$\pi(\phi | X_e, X_o, \eta) \sim \mathcal{N} \left\{ \hat{\phi}_\eta, (n_e V_e^{-1} + \eta n_o V_o^{-1})^{-1} \right\}, \quad (22)$$

where $\hat{\phi}_\eta = (n_e V_e^{-1} + \eta n_o V_o^{-1})^{-1} (n_e V_e^{-1} \hat{\phi}_e + \eta n_o V_o^{-1} \hat{\phi}_o) \in \mathbb{R}^P$, with $\hat{\phi}_s \in \mathbb{R}^P$ and $V_s \in \mathbb{R}^{P \times P}$ denoting separate maximum likelihood estimates and asymptotic sandwich covariance matrices for source $s \in \{e, o\}$.

Exponentiation by η scales the observational precision by ηn_o . Beyond theoretical elegance, (22) provides substantial operational advantages. Computationally, evaluating the Gaussian density in closed form enables rapid, Markov chain Monte Carlo-free grid evaluations of $\hat{\eta}$, bypassing the computational bottlenecks of high-dimensional sampling across large observational registries. From a governance perspective, this asymptotic decoupling facilitates privacy-preserving multi-centre data fusion, as participating sites need only exchange summary estimates and sandwich covariance matrices $(\hat{\phi}_e, \hat{\phi}_o, V_e, V_o)$ rather than pooling individual patient-level data.

Given the optimal $\hat{\eta}$, posterior draws $\theta^{(m)} \sim \pi(\theta | X_e, X_o, \hat{\eta})$ with $\theta^{(m)} \in \mathbb{R}^{p_\theta}$ yield the average treatment effect by empirical g-computation over the trial cohort:

$$\hat{\tau}^{(m)} = \frac{1}{n_e} \sum_{i \in D_e} \left[E \{ Y(1) | C_i; \theta^{(m)} \} - E \{ Y(0) | C_i; \theta^{(m)} \} \right], \quad (23)$$

from which posterior point estimates and credible intervals follow directly.

The combination of power likelihood and frugal parameterisation provides an adaptable framework for clinical trial data fusion. Unlike contrast-matching estimators that require both treatment

arms to be observed across all data sources, this formulation accommodates both single-arm (control-only) and two-arm observational data, yielding substantial precision gains under causal concordance. Moreover, in the absence of unmeasured confounding, simulations demonstrate that data-adaptive power likelihood estimation achieves substantial variance reduction, narrowing posterior credible interval widths by up to 45% and closely approaching the theoretical oracle benchmark (Lin et al., 2026). 305

Nonetheless, the approach exhibits a *risk–reward* trade-off. Because $\hat{\eta}$ is determined by optimising the omnibus joint predictive density over all variables X rather than isolating the treatment contrast $\tau(c)$, moderate unmeasured confounding may fail to trigger sufficient predictive penalties. In this intermediate regime, the posterior can suffer from transient credible interval under-coverage before severe bias forces $\hat{\eta} \rightarrow 0$. Addressing this risk requires sensitivity analyses and careful pre-specification of the causal graphical structure. 310

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